

Lesson 4: Gyro Sensors

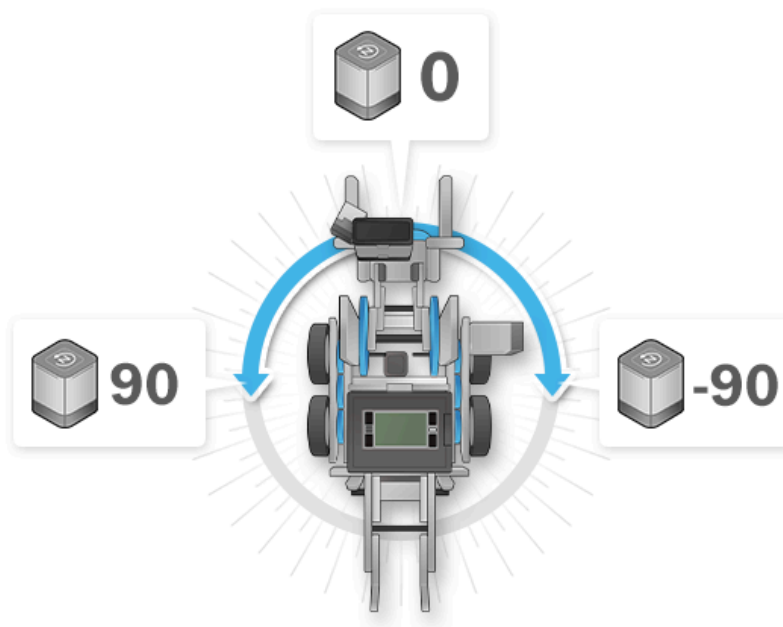
What is the Gyro Sensor?



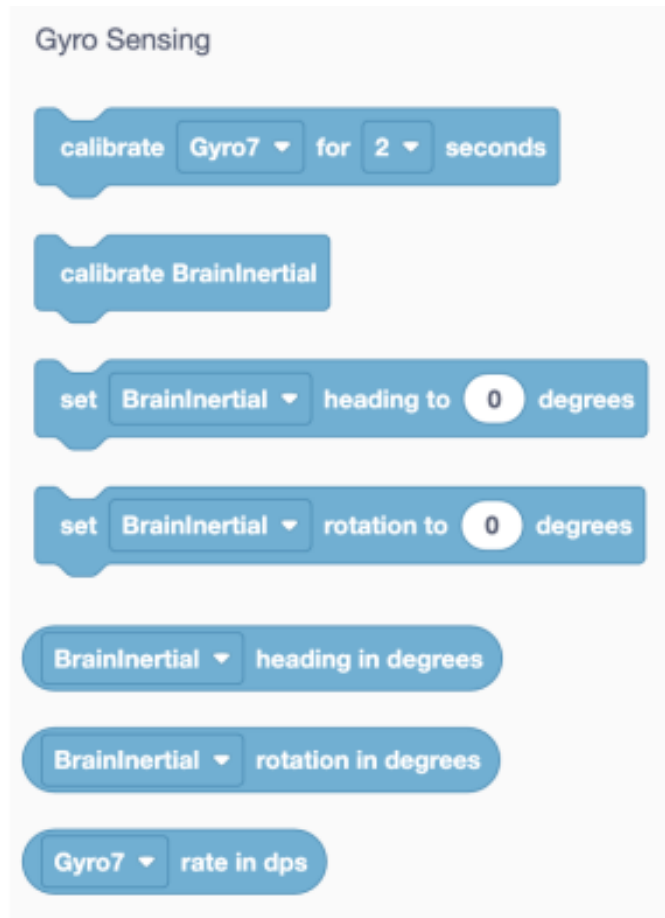
The VEX IQ Gyro Sensor helps your robot measure how much it turns (in degrees). It's like a compass for your robot.

It measures the rate of rotation and helps the robot find the direction it is facing in degrees. It is useful for:

- Making precise turns and maintaining direction.
- Correcting a robot's direction if it strays from a straight path.
- Running smooth and controlled navigation.



Gyro Sensing Code Blocks



1. Initialize the Gyro Sensor

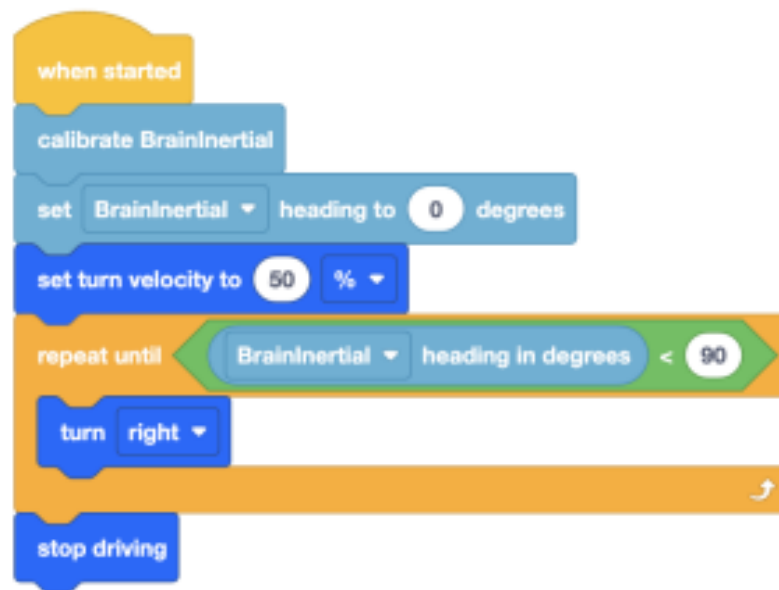
It's important to reset or "calibrate" the Gyro Sensor before making any turns. The **[Calibrate Gyro Sensor]** block resets the gyro reading to 0 degrees before starting the turn. Place **[Calibrate Gyro Sensor]** at the start of the program to ensure it starts from 0 degrees.

2. 90° Turns:

Add the **[Repeat Until]** block. Set the block's condition to "**Gyro Sensor angle < 90 degrees**". This will keep checking the angle until the robot has turned exactly 90 degrees.

Inside the **[Repeat Until]** block, add **[Turn Right]** or **[Turn Left]** block to turn.

After the turn is complete, add the **[Stop]** block to stop the drivetrain.



Explain the code structure in your own words, as if you were explaining to a friend with no robotics experience.

Testing Your Program

Once the code is complete, run the program. Your robot should make a precise 90-degree turn based on the Gyro Sensor's feedback.

Exercise 1: Driving in a Square

Program the robot to drive in a square shape using the Gyro Sensor for controlled 90-degree turns at each corner. Submit a screenshot of your code to the corresponding Google Classroom assignment.

Exercise 2: Straight-Line Driving

Program the robot to drive in a straight line using the Gyro Sensor to correct any misleading direction. Submit a screenshot of your code to the corresponding Google Classroom assignment.

